

Hassaan Ahmed

Senior Full Stack Engineer

hassaan.ahmed.work@gmail.com | +92-314-5250088 | Islamabad, Pakistan, 46000 | hassaanahmed.dev | github.com/murtrax
linkedin.com/in/hassaan-ahmed-293ba818a/

SUMMARY

Senior Full Stack Engineer with 7+ years of experience building scalable web applications, AI-driven workflow systems, and decentralized applications. Strong expertise in TypeScript/JavaScript ecosystems, startup-scale architecture, and remote-first collaboration with distributed teams. Proven track record of owning production systems end-to-end and shipping high-performance solutions in fast-paced startup environments.

WORK EXPERIENCE

- K3-Labs, Senior Full Stack Engineer**

03/2024 – Present | New York, USA | Remote

 - Joined as a founding engineer and engineered the architecture and MVP before the company's initial funding round.
 - Architected a large-scale multi-agent AI system enabling natural-language creation and deployment of complex automation workflows with deterministic structure, dependency, and intent resolution.
 - Implemented AI workflow automation system with 30+ type-safe functions and full TypeScript inference, enabling LangGraph agents, MCP server integrations, and multi-agent orchestration.
 - Shipped backend infrastructure powering thousands of concurrent AI workflows with event-driven execution, Redis-based state management, and fault-tolerant orchestration across horizontally scaled services.
 - Deployed a decentralised execution layer operated by independent nodes, providing secure task isolation, peer-to-peer communication, automatic HTTPS tunneling, and integrated monitoring and alerting via Prometheus and Grafana.
- Team.Finance, Full Stack Engineer**

06/2022 – 03/2024 | Canada | Remote

 - Architected blockchain indexing pipelines, improving query performance by 72% and eliminating third-party data provider dependencies to power real-time DeFi dashboards.
 - Optimized NestJS full-stack web application performance by 50%+, reducing LCP, FCP, and TTI through code splitting, lazy loading, and API optimization strategies.
 - Led migration from a monolithic NestJS backend to a microservice architecture, introducing Pub/Sub-driven event communication to decouple services, improve fault isolation, and support independent scaling of high-throughput workloads.
 - Refactored NestJS backend with stricter TypeScript practices, enhanced error handling, and type-safe patterns, reducing production error rates by 30% and improving code maintainability.
 - Migrated multiple frontend applications to monorepo architecture, establishing shared component libraries and build pipelines that boosted development velocity and code consistency across independent verticals.
 - Promoted to Team Lead for 5-engineer team, accelerating feature delivery by 40% through improved sprint planning, code review processes, and clear technical specifications from business requirements.
- AKRU, Full Stack Engineer**

07/2019 – 06/2022 | Islamabad, Pakistan | Onsite

 - Created a full-stack real estate tokenization platform enabling SEC-compliant fractional ownership, primary and secondary sales/transfers, and end-to-end property lifecycle management for \$10M+ in tokenized properties.
 - Built AWS HSM-backed EVM wallet solution replacing external providers like Magic.link, cutting authentication costs by 50% and enabling white-label wallet-as-a-service offerings to clients.
 - Owned scalable backend systems and blockchain indexing pipelines delivering real-time dashboards, reducing API latency from seconds to milliseconds through query optimization and caching strategies.

EDUCATION

- Computer Science, National University of Sciences & Technology (NUST)**

09/2016 – 07/2020 | Islamabad, Pakistan

 - GPA: 3.5

SKILLS

Languages TypeScript, JavaScript	Frontend React, Next.js, TailwindCSS, React Flow, TanStack, HTML/CSS, Redux	Backend Node.js, NestJS, PostgreSQL, Redis, Prisma, REST APIs, GraphQL, Socket.IO, Pub/Sub, Prometheus/Grafana	AI/ML LangChain, LangGraph, Vector databases, RAG, MCP, OpenAI API, Anthropic API, Embeddings
DevOps Docker, Git, AWS, GCP, CI/CD, GitHub Actions	Practices Test-Driven Development, Agile/Scrum, Code Review, System Design, Microservices, Event-driven Architecture		